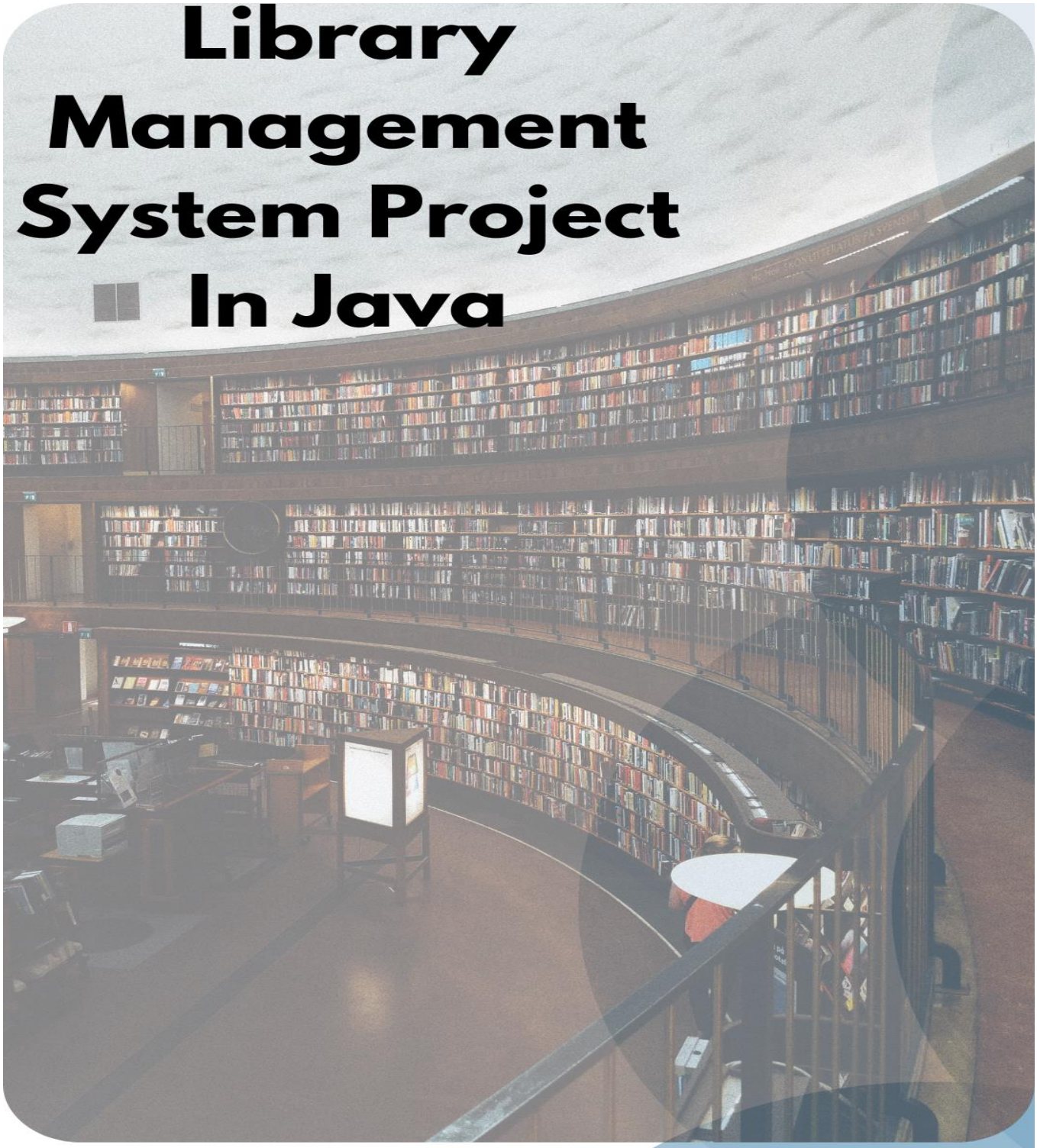
Library Management System Project In Java



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Project Report On

LIBRARY MANAGEMENT SYSTEM

Submitted By:

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BCA DEGREE

In

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ABSTRACT

Library management system is a project which aims in developing a computerized system to maintain all the daily work of library .This project has many features which are generally not available in normal library management systems like facility of user login and a facility of teachers login .It also has a facility of admin login through which the admin can monitor the whole system .It also has facility of an online notice board where teachers can student can put up information about workshops or seminars being held in our colleges or nearby colleges and librarian after proper verification from the concerned institution organizing the seminar can add it to the notice board . It also has a facility where students, after logging in their accounts, can see list of books issued and its issue date and return date and the students can request the librarian to add new books by filling the book request form. The librarian, after logging into his account i.e., admin account can generate various reports such as student report, issue report, teacher report and book report.

Overall, this project of ours is being developed to help the students as well as the staff of the library to maintain the library in the best way possible and also reduce the human efforts.

INTRODUCTION

This introduction provides an overview of the purpose, objectives, background and operating environment of the system.

1.1 PROJECT AIMS AND OBJECTIVES

The project aims and objectives that will be achieved after completion of this project are discussed in this subchapter. The aims and objectives are as follows:

- Book issue
- Requires column for librarian to provide new books.
- A separate column for digital library
- Admin login page where admin can find books issued by him/her and date of return.
- A search column to search availability of books
- Book table for book details

1.2 BACKGROUND OF PROJECT

Library Management System is an application which refers to library systems which are generally small or medium in size. It is used by the librarian to manage the library using a computerized system where he/she can record various transactions like issue of books, return of books, addition of new books, addition of new students etc.

Books and student maintenance modules are also included in this system which would keep track of the students using the library and a detailed description of the books a library contains. With this computerized system there will be no loss of book record or member record which generally happens when a non-computerized system is used.

In addition, report modules are also included in the Library Management System. If the user's position is admin, the user can generate different kinds of reports like lists of students registered, listof books, issue and return reports.

All these modules can help librarians to manage the library with more convenience and ina more efficient way as compared to library systems which are not computerized.

1.3 OPERATION ENVIRONMENT

PROCESSOR	INTEL CORE PROCESSOR OR BETTER PERFORMANCE
OPERATING SYSTEM	WINDOWS VISTA ,WINDOWS7, UBUNTU
MEMORY	1GB RAM OR MORE
HARD DISK SPACE	MINIMUM 3 GB FOR DATABASE USAGE FOR FUTURE
FRONT END	Java SE 18.0. 2.1
DATABASE	MY SQL

SYSTEM ANALYSIS

In this chapter, we will discuss and analyze the developing process of Library Management System including software requirement specification (SRS) and comparison between existing and proposed system. The functional and non-functional requirements are included in SRS part to provide complete description and overview of system requirement before the developing process is carried out. Besides that, existing vs proposed provides a view of how the proposed system will be more efficient than the existing one.

2.1 SOFTWARE REQUIREMENT SPECIFICATION

2.1.1 GENERAL DESCRIPTION

PRODUCT DESCRIPTION:

Library Management System is a computerized system which helps user(librarian) to manage the library daily activity in electronic format. It reduces the risk of paperwork such as file lost, file damaged and time consuming.

It can help the user to manage the transaction or record more effectively and time-saving.

PROBLEM STATEMENT:

The problem occurred before having computerized system includes:

- File lost.

  When a computerized system is not implemented files are always lost because of the human environment. Sometimes due to some human error there may be a loss of records.

- File damaged

  When a computerized system is not there, the file is always lost due to some accident like spilling of water by some member on file accidentally. Besides, some natural disasters like floods or fires may also damage the files.

- Difficult to search record

  When there is no computerized system there is always a difficulty in searching of records if the records are large in number.

- Space consuming

  After the number of records become large the space for physical storage of file and records also increases if no computerized system is implemented.

- Cost consuming

  As there is no computerized system the to add each record paper will be needed which will increase the cost for the management of library.

2.1.2 SYSTEM OBJECTIVES

- Improvement in control and performance

  The system is developed to cope up with the current issues and problems of library . The system can add user, validate user and is also bug free.

- Save cost

  After computerized system is implemented less human force will be required to maintain the library thus reducing the overall cost.

- Save time

  Librarian is able to search record by using few clicks of mouse and few search keywords thus saving his valuable time.

2.1.3 SYSTEM REQUIREMENTS

2.1.3.1 NON-FUNCTIONAL REQUIREMENTS

- Product Requirements

EFFICIENCY REQUIREMENT

When a library management system will be implemented librarian and user will easily acess library as searching and book transaction will be very faster .

RELIABILITY REQUIREMENT

The system should accurately performs member registration ,member validation , report generation, book transaction and search

USABILITY REQUIREMENT

The system is designed for a user friendly environment so that student and staff of library can perform the various tasks easily and in an effective way.

ORGANIZATIONAL REQUIREMENT

IMPLEMENTATION REQUIREMNTS

In implementing whole system it uses html in front end with php as server side scripting language which will be used for database connectivity and the backend ie the database part is developed using mysql.

DELIVERY REQUIREMENTS

The whole system is expected to be delivered in six months of time with a weekly evaluation by the project guide.

2.1.3.2 FUNCTIONAL REQUIREMENTS

1. NORMAL USER

1.1 USER LOGIN

Description of feature

This feature used by the user to login into system. They are required to enter user id and password before they are allowed to enter the system. The user id and password will be verified and if invalid id is there user is allowed to not enter the system.

Functional requirements

-user id is provided when they register

-The system must only allow user with valid id and password to enter the system

-The system performs authorization process which decides what user level can access to.

-The user must be able to logout after they finished using system.

1.2 REGISTER NEW USER

Description of feature

This feature can be performed by all users to register new user to create account.

Functional requirements

-System must be able to verify information

-System must be able to delete information if information is wrong

1.3 REGISTER NEW BOOK

Description of feature

This feature allows to add new books to the library

Functional requirements

-System must be able to verify information

-System must be able to enter number of copies into table.

- System must be able to not allow two books having same book id.

1.5 SEARCH BOOK

DESCRIPTION OF FEATURE

This feature is found in book maintenance part . we can search book based on book id , book name , publication or by author name.

Functional requirements

- System must be able to search the database based on select search type
- System must be able to filter book based on keyword enterd
- System must be able to show the filtered book in table view

1.5 ISSUE BOOKS AND RETURN BOOKS

DESCRIPTION OF FEATURE

This feature allows to issue and return books and also view reports of book issued.

Functional requirements

-System must be able to enter issue information in database.

-System must be able to update number of books.

- System must be able to search if book is available or not before issuing books

-System should be able to enter issue and return date information

.

2.1.4 SOFTWARE AND HARDWARE REQUIREMENTS

This section describes the software and hardware requirements of the system

2.1.4.1 SOFTWARE REQUIREMENTS

- Operating system- Windows 7 is used as the operating system as it is stable and supports more features and is more user friendly
- Database MYSQL is used as database as it easy to maintain and retrieverecords by simple queries which are in English language which are easy to understand and easy to write.
- Development tools and Programming language- JAVA is used to write the whole code.

2.1.4.2 HARDWARE REQUIREMENTS

- Intel core i5 2nd generation is used as a processor because it is fast than other processors an provide reliable and stable and we can run our pc for longtime. By using this processor we can keep on developing our project without any worries.
- Ram 1 gb is used as it will provide fast reading and writing capabilities and will in turn support in processing

2.2 EXISTING VS PROPOSED SYSTEM

i. Existing system does not have any facility of admin login whereas proposed system will have a facility of admin login

ii. Existing system does not have a facility of online reservation of books whereas proposed system has a facility of reservation of books

iii. Existing system does not have any facility to generate student reports as well book issue reports whereas proposed system provides librarian with a tool to generate reports

iv. Existing system does not has any facility for book request and suggestions where as in proposed system after logging in to their accounts student can request books as well as provide suggestions to improve library

2.3 SOFTWARE TOOLS USED

The whole Project is divided in two parts the front end and the back end.

2.3.1 Front end

The front end is designed using of JAVA

- JAVA- **Java** is a high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible. It is a general-purpose programming language intended to let programmers *write once, run anywhere* (WORA), meaning that compiled Java code can run on all platforms that support Java without the need to recompile. Java applications are typically compiled to bytecode that can run on any Java virtual machine (JVM) regardless of the underlying computer architecture. The syntax of Java is similar to C and C++, but has fewer low-level facilities than either of them. The Java runtime provides dynamic capabilities (such as reflection and runtime code modification) that are typically not available in traditional compiled languages. As of 2019, Java was one of the most popular programming languages in use according to GitHub, particularly for client–server web applications, with a reported 9 million developers.CSS-

2.3.2 BACK END- The back end is designed using mysql which is used to design the databases

- MYSQL- **MySQL** ("My S-Q-L", officially, but also called "My Sequel") is (as of July 2013) the world's second most widely used open-source relational database management system (RDBMS). It is named after co-founder Michael Widenius daughter, My. The SQL phrase stands for Structured Query Language. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation .MySQL is a popular choice of database for use in web applications, and is a central component of the widely used LAMP open source web application software stack (and other 'AMP' stacks). LAMP is an acronym for "Linux, Apache, MySQL, Perl/PHP/Python." Free-software-open source projects that require a full-featured database management system often use MySQL. For commercial use, several paid editions are available, and offer additional functionality. Applications which use MySQL databases

SYSTEM DESIGN

3.1 TABLE DESIGN

VARIOUS TABELS TO MAINTAIN INFORMATION

Create a database

```
-- Creating database
create database library;
```

Select the database

```
-- Selecting the database
use library;
```

Create a book table

```
Creating book table
create table book (
  id int primary key,
  book_name varchar(25),
  category varchar(15),
  author varchar(25),
  publisher varchar(25),
);
```

Create an issue table

```
-- Creating issue table
create table issue (
  member_id int primary key,
  member_name varchar(25),
  book_name varchar(25),
  date_issue date,
  date_return date
);
```

Create an admin table

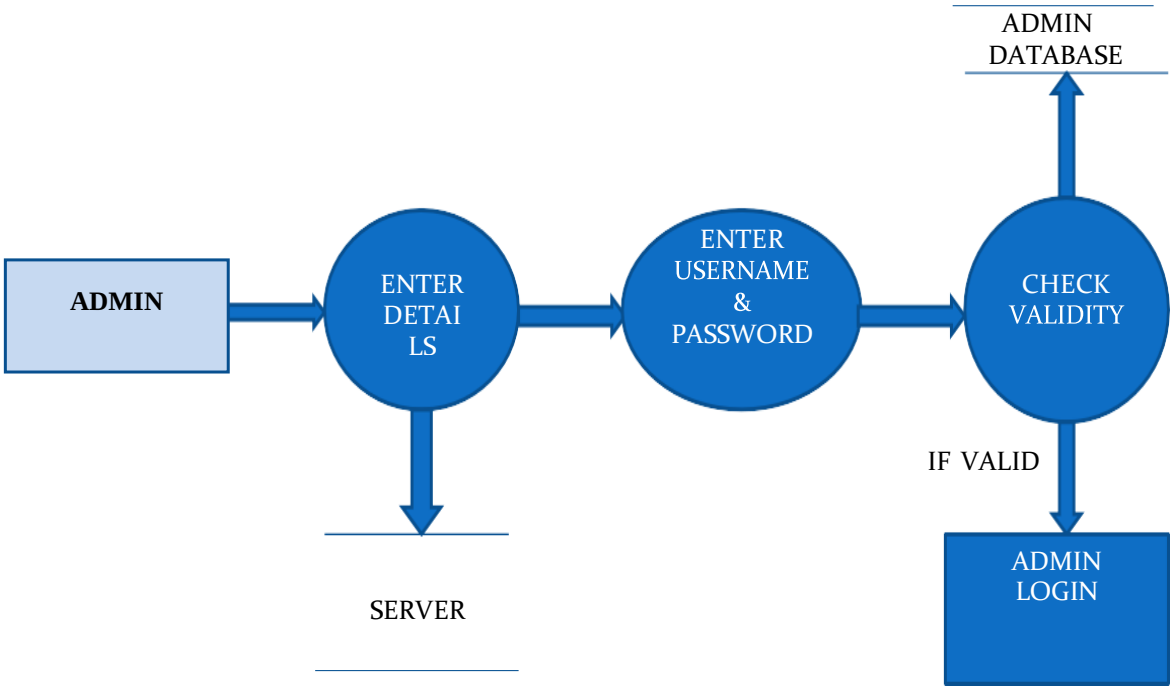
```
-- Create admin table
create table admin (
cust_id varchar(25),
password varchar(25)
);
```

Insert some values in the admin table

```
-- Inserting values
insert into admin values
("admin1","1234"),
("admin2","9876");
```

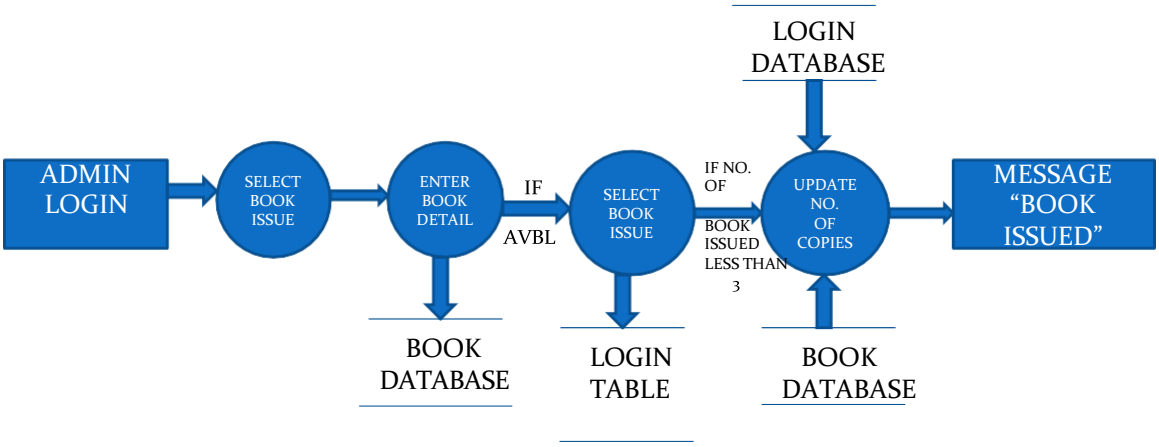
3.2 DATA FLOW DIAGRAMS

DATA FLOW DIAGRAM FOR TEACHER LOGIN



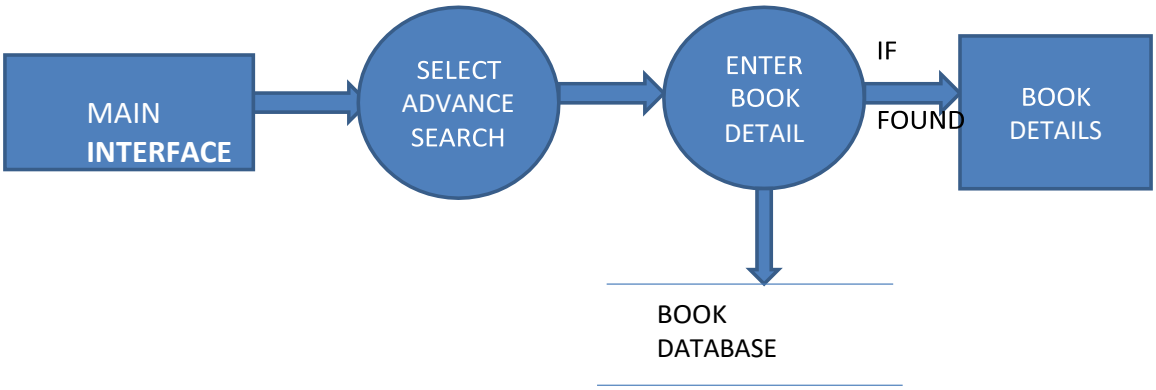
After entering to the application, admin can choose the ADMIN LOGIN option where they are asked to enter username & password , and if he/she is a valid user thena ADMIN login INTERFACE will be displayed.

DATA FLOW DIAGRAM FOR BOOK ISSUE



It is a 2nd level Data Flow Diagram where after entering ADMIN LOGIN INTERFACE he/she can select a book issue option where after entering the book detail, he/she can select the book issue option and if the maximum no of books issued limit is not crossed then the book will be issueed.

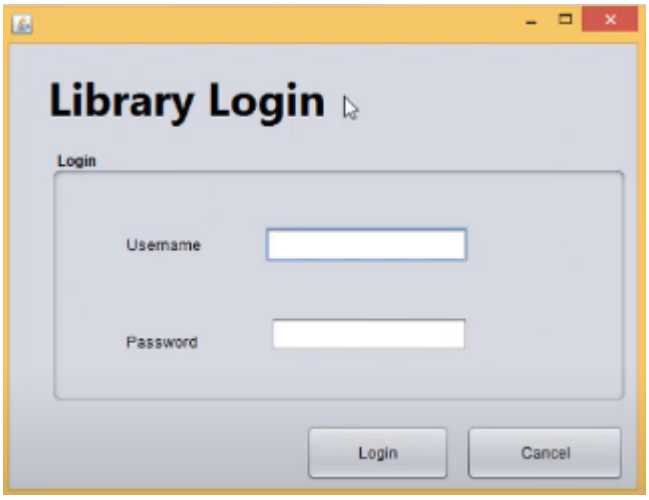
DATA FLOW DIAGRAM FOR BOOK SEARCH



After the home page login there will be an option of the book search where after entering book detail like author name, publication, book name etc book details will be displayed.

SYSTEM IMPLEMENTATION

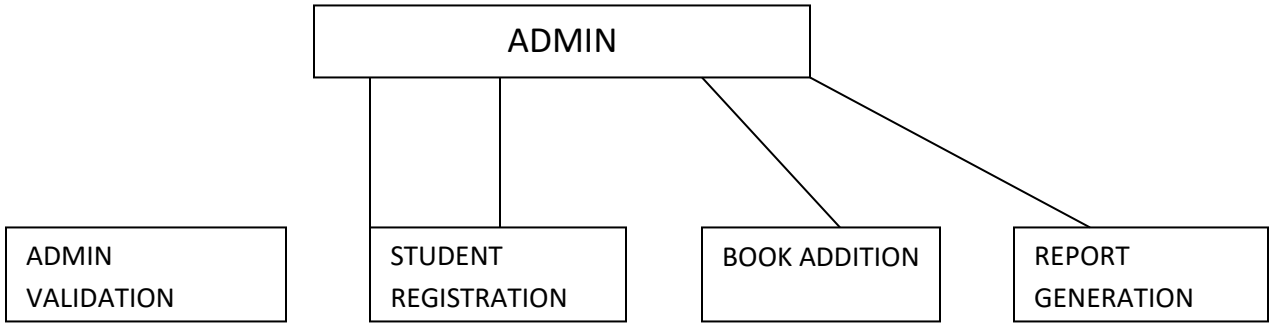
4.1.1 Screenshot for Login



4.1 MODULE DESCRIPTION

For Library Management System it is divided into the following Modules:

4.1.1 Admin Module



Login module in library management system project in java

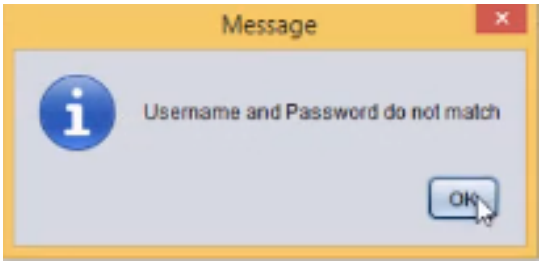
This module helps to make a login page so that the user can enter the user name and password to enter into the library management system in java. Name the file **login.java**

```
package library;
import javax.swing.JOptionPane;
public class Login extends javax.swing.JFrame {
  public Login() {
    initComponents();
  }
  @SuppressWarnings("unchecked")
  private void initComponents() {
    jLabel1 = new javax.swing.JLabel();
    jPanel1 = new javax.swing.JPanel();
    jLabel2 = new javax.swing.JLabel();
    jLabel3 = new javax.swing.JLabel();
    txtusername = new javax.swing.JTextField();
    txtpass = new javax.swing.JPasswordField();
    jButton2 = new javax.swing.JButton();
    jButton1 = new javax.swing.JButton();
    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 36));
    jLabel1.setText("Library Login");
    jPanel1.setBorder(javax.swing.BorderFactory.createTitledBorder("Login"));
    jLabel2.setText("Username");
    jLabel3.setText("Password");
    javax.swing.GroupLayout jPanel1Layout = new javax.swing.GroupLayout(jPanel1);
    jPanel1.setLayout(jPanel1Layout);
    jPanel1Layout.setHorizontalGroup(
      jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
      .addGroup(jPanel1Layout.createSequentialGroup()
        .addGap(48, 48, 48)
        .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
          .addComponent(jLabel2)
```

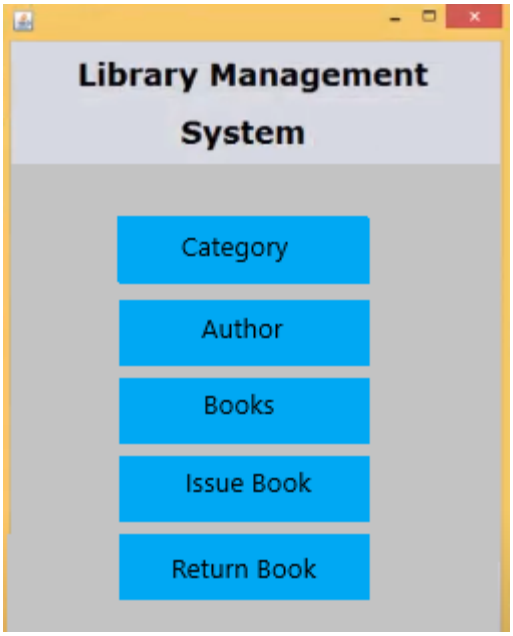
```
                .addComponent(jLabel3))
            .addGap(51, 51, 51)
            .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING, false)
                .addGroup(jPanel1Layout.createSequentialGroup()
                    .addGap(4, 4, 4)
                    .addComponent(txtpass))
                .addComponent(txtusername, javax.swing.GroupLayout.PREFERRED_SIZE, 161,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addContainerGap(114, Short.MAX_VALUE))
        );
        jPanel1Layout.setVerticalGroup(
            jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, jPanel1Layout.createSequentialGroup()
                .addGap(33, 33, 33)
                .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(jLabel2)
                    .addComponent(txtusername, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(43, 43, 43)
                .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING)
                    .addComponent(jLabel3)
                    .addComponent(txtpass, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addContainerGap(31, Short.MAX_VALUE))
        );
        jButton2.setText("Cancel");
        jButton2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton2ActionPerformed(evt);
            }
        });
        jButton1.setText("Login");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(31, 31, 31)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(jLabel1)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING)
                        .addGroup(layout.createSequentialGroup()
                            .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 115,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                            .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 126,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addGap(4, 4, 4))
```

```
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addContainerGap(18, Short.MAX_VALUE)
            .addComponent(jLabel1)
            .addGap(18, 18, 18)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
                    .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(64, 64, 64))
                .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 45,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 45,
javax.swing.GroupLayout.PREFERRED_SIZE))
                    .addContainerGap()))
    );
    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton1ActionPerformed
    String username = txtusername.getText();
    String pass = txtpass.getText();
    if(username.equals("Peter") && pass.equals("123"))
    {
      Main m = new Main();
       this.hide();
       m.setVisible(true);
    }
    else
    {
       JOptionPane.showMessageDialog(this,"Username and Password do not match");
       txtusername.setText("");
        txtpass.setText("");
        txtusername.requestFocus();
    }
  }
  }
```

When the entered user name and password didn't match the existing records in the database, it will return the "Username and password do not match".



Once we enter the correct username and password it takes us to the home page of our library management application.



2. Category Module in library management system in java

We can create a new category in this module by giving the details. Name the file **category.java**.

```
package library;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.SQLException;
import java.util.Vector;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;
public class category extends javax.swing.JFrame {
    public category() {
        initComponents();
        Connect();
        table_update();
    }
Connection con;
PreparedStatement pst;
     private void initComponents() {
        jPanel2 = new javax.swing.JPanel();
        jLabel9 = new javax.swing.JLabel();
        jLabel10 = new javax.swing.JLabel();
        txtcat = new javax.swing.JTextField();
        txtstatus = new javax.swing.JComboBox<>();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jButton3 = new javax.swing.JButton();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTable1 = new javax.swing.JTable();
        jButton4 = new javax.swiew javax.swing.JButton();
        jLabel1 = new javax.swing.JLabel();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jPanel2.setBorder(javax.swing.BorderFactory.createTitledBorder(null, "Category",
javax.swing.border.TitledBorder.DEFAULT_JUSTIFICATION,
javax.swing.border.TitledBorder.DEFAULT_POSITION, new java.awt.Font("Tahoma", 1, 14)));
        jLabel9.setText("Category");
        jLabel10.setText("Status");
        txtstatus.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "Active", "DeActive" }));
        jButton1.setText("Add");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
        jButton2.setText("Edit");
        jButton2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton2ActionPerformed(evt);
            }
```

```
        });

        jButton3.setText("Delete");
        jButton3.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton3ActionPerformed(evt);
            }
        });
        jTable1.setModel(new javax.swing.table.DefaultTableModel(
            new Object [][] {
            },
            new String [] {
                "Id", "Category", "Status"
            }
        ) {
            Class[] types = new Class [] {
                java.lang.Integer.class, java.lang.String.class, java.lang.String.class
            };
            public Class getColumnClass(int columnIndex) {
                return types [columnIndex];
            }
        });
        jTable1.setCursor(new java.awt.Cursor(java.awt.Cursor.DEFAULT_CURSOR));
        jTable1.setGridColor(new java.awt.Color(255, 255, 255));
        jTable1.addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseClicked(java.awt.event.MouseEvent evt) {
                        jTable1MouseClicked(evt);
            }
        });
        jScrollPane1.setViewportView(jTable1);
        jButton4.setText("Cancel");
        jButton4.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton4ActionPerformed(evt);
            }
        });

        javax.swing.GroupLayout jPanel2Layout = new javax.swing.GroupLayout(jPanel2);
        jPanel2.setLayout(jPanel2Layout);
        jPanel2Layout.setHorizontalGroup(
            jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(26, 26, 26)
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING)
                    .addGroup(jPanel2Layout.createSequentialGroup()
                        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(jLabel9)
                            .addComponent(jLabel10))
                        .addGap(35, 35, 35)
                        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(txtstatus, javax.swing.GroupLayout.PREFERRED_SIZE, 153,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
                    .addComponent(txtcat, javax.swing.GroupLayout.PREFERRED_SIZE, 153,
javax.swing.GroupLayout.PREFERRED_SIZE)))
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 75,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .addGroup(jPanel2Layout.createSequentialGroup()
                            .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 73,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addGap(18, 18, 18)
                            .addComponent(jButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 83,
javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addComponent(jButton4, javax.swing.GroupLayout.PREFERRED_SIZE, 104,
javax.swing.GroupLayout.PREFERRED_SIZE))
                    .addGap(8, 8, 8)))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 73, Short.MAX_VALUE)
            .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 427,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap())
        );


    jPanel2Layout.setVerticalGroup(
            jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(21, 21, 21)
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(txtcat, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jLabel9))
                .addGap(27, 27, 27)
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(txtstatus, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jLabel10))
                .addGap(44, 44, 44)
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 39,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 39,
javax.swing.GroupLayout.PREFERRED_SIZE))
                    .addComponent(jButton3, javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.PREFERRED_SIZE, 39, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(32, 32, 32)
                .addComponent(jButton4, javax.swing.GroupLayout.PREFERRED_SIZE, 42,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 260,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
            .addGap(0, 30, Short.MAX_VALUE))
      );

    jLabel1.setFont(new java.awt.Font("Tahoma", 1, 36));
       javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
       getContentPane().setLayout(layout);
       layout.setHorizontalGroup(
          layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
          .addGroup(layout.createSequentialGroup()
             .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                   .addContainerGap()
                   .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGroup(layout.createSequentialGroup()
                   .addGap(287, 287, 287)
                   .addComponent(jLabel1)))
             .addContainerGap(23, Short.MAX_VALUE))
       );
       layout.setVerticalGroup(
          layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
          .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
             .addGap(27, 27, 27)
             .addComponent(jLabel1)
             .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 43, Short.MAX_VALUE)
             .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
             .addContainerGap())
       );

  public void Connect()
    {
       try {
          Class.forName("com.mysql.jdbc.Driver");
            con = DriverManager.getConnection("jdbc:mysql://localhost/slibr","root","");
       } catch (ClassNotFoundException ex) {
          Logger.getLogger(category.class.getName()).log(Level.SEVERE, null, ex);
       } catch (SQLException ex) {
          Logger.getLogger(category.class.getName()).log(Level.SEVERE, null, ex);
       }
     private void table_update()
    {
          int c;
          try {
              pst = con.prepareStatement("select * from category");
              ResultSet rs = pst.executeQuery();
              ResultSetMetaData rsd = rs.getMetaData();
              c = rsd.getColumnCount()
              DefaultTableModel d = (DefaultTableModel)jTable1.getModel();
              d.setRowCount(0);
              while(rs.next())
              {
```

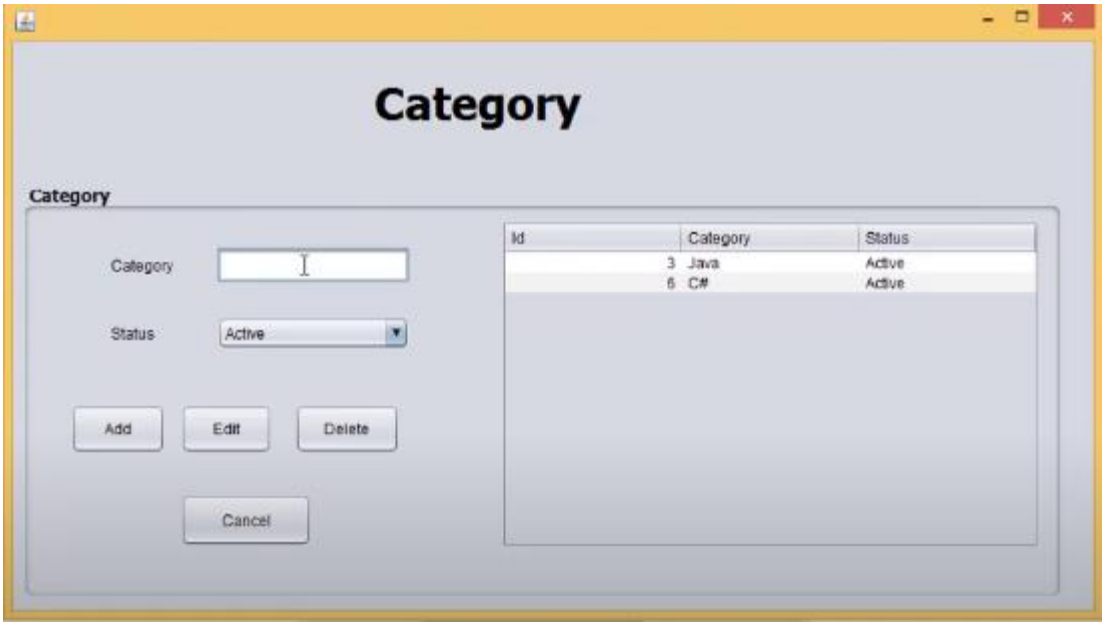
```
                Vector v2 = new Vector();
                for(int i=1; i<=c; i++)
                {
                   v2.add(rs.getString("id"));
                   v2.add(rs.getString("category"));
                   v2.add(rs.getString("status"));
                }
                d.addRow(v2);
             }
        } catch (SQLException ex) {
           Logger.getLogger(category.class.getName()).log(Level.SEVERE, null, ex);
        }
     }

  private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
        DefaultTableModel d1 = (DefaultTableModel)jTable1.getModel();
        int selectIndex = jTable1.getSelectedRow();
        int id = Integer.parseInt(d1.getValueAt(selectIndex, 0).toString());
        String category = txtcat.getText();
        String status = txtstatus.getSelectedItem().toString();
        try {
           pst = con.prepareStatement("update category set category=?,status=? where id= ?");
           pst.setString(1, category);
           pst.setString(2, status);
           pst.setInt(3, id);
           pst.executeUpdate();
           JOptionPane.showMessageDialog(null,"Category Updateddd");
           table_update();
           txtcat.setText("");
           txtstatus.setSelectedIndex(-1);
           txtcat.requestFocus();
             jButton1.setEnabled(true);
        }
        catch (SQLException ex) {
           Logger.getLogger(category.class.getName()).log(Level.SEVERE, null, ex);
        }

  private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton3ActionPerformed
        DefaultTableModel d1 = (DefaultTableModel)jTable1.getModel();
        int selectIndex = jTable1.getSelectedRow();
        int id = Integer.parseInt(d1.getValueAt(selectIndex, 0).toString());
        int dialogResult = JOptionPane.showConfirmDialog(null, "Do you want to Delete the
Record","Warning",JOptionPane.YES_NO_OPTION);
        if(dialogResult == JOptionPane.YES_OPTION)
        {
           try {
              pst = con.prepareStatement("delete from category where id =?");
                pst.setInt(1, id);
              pst.executeUpdate();
              JOptionPane.showMessageDialog(null,"Category Deletedd");
              table_update();
```

```
            txtcat.setText("");
            txtstatus.setSelectedIndex(-1);
            txtcat.requestFocus();
        } catch (SQLException ex) {
            Logger.getLogger(category.class.getName()).log(Level.SEVERE, null, ex);
        }
 private void jTable1MouseClicked(java.awt.event.MouseEvent evt) {
      DefaultTableModel d1 = (DefaultTableModel)jTable1.getModel();
       int selectIndex = jTable1.getSelectedRow();
       txtcat.setText(d1.getValueAt(selectIndex, 1).toString());
       txtstatus.setSelectedItem(d1.getValueAt(selectIndex, 2).toString());
       jButton1.setEnabled(false);
}

  private void jButton4ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton4ActionPerformed
      this.setVisible(false);
   }
   public static void main(String args[]) {
     try {
         for (javax.swing.UIManager.LookAndFeelInfo info : javax.swing.UIManager.getInstalledLookAndFeels())
{
            if ("Nimbus".equals(info.getName())) {
               javax.swing.UIManager.setLookAndFeel(info.getClassName());
               break;
            }
         }
       } catch (ClassNotFoundException ex) {
          java.util.logging.Logger.getLogger(category.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
       } catch (InstantiationException ex) {
          java.util.logging.Logger.getLogger(category.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
       } catch (IllegalAccessException ex) {
          java.util.logging.Logger.getLogger(category.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
       } catch (javax.swing.UnsupportedLookAndFeelException ex) {
          java.util.logging.Logger.getLogger(category.class.getName()).log(java.util.logging.Level.SEVERE, null,
ex);
      java.awt.EventQueue.invokeLater(new Runnable() {
          public void run() {
             new category().setVisible(true);
          }
       });
)
```



3. Author module in library management system project,we can create new author profiles. Name the file as **author.java**.

```
package library;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.SQLException;
import java.util.Vector;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;
public class author extends javax.swing.JFrame {
    public author() {
        initComponents();
```

```
        Connect();
        table_update();
    }
Connection con;
PreparedStatement pst;
    @SuppressWarnings("unchecked")
    private void initComponents() {
        jToggleButton1 = new javax.swing.JToggleButton();
        jPanel2 = new javax.swing.JPanel();
        jLabel9 = new javax.swing.JLabel();
        txtname = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jButton3 = new javax.swing.JButton();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTable1 = new javax.swing.JTable();
        jLabel11 = new javax.swing.JLabel();
        jLabel12 = new javax.swing.JLabel();
        txtphone = new javax.swing.JTextField();
        jScrollPane2 = new javax.swing.JScrollPane();
        txtaddress = new javax.swing.JTextArea();
        jButton4 = new javax.swing.JButton();
        jLabel1 = new javax.swing.JLabel();
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
     jButton2.setText("Edit");
        jButton2.addActionListener(new java.awt.event.ActionListener() {
```

```
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton2ActionPerformed(evt);
        }
    });
    jButton3.setText("Delete");
    jButton3.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton3ActionPerformed(evt);
        }
    });
    jTable1.setModel(new javax.swing.table.DefaultTableModel(
        new Object [][] {
        },
        new String [] {
            "Id", "Name", "Address", "Phone"
        }
    ) {
        Class[] types = new Class [] {
            java.lang.Integer.class, java.lang.String.class, java.lang.String.class,
java.lang.String.class
        };
        public Class getColumnClass(int columnIndex) {
            return types [columnIndex];
        }
    });
 jTable1.setCursor(new java.awt.Cursor(java.awt.Cursor.DEFAULT_CURSOR));
    jTable1.setGridColor(new java.awt.Color(255, 255, 255));
    jTable1.addMouseListener(new java.awt.event.MouseAdapter() {
        public void mouseClicked(java.awt.event.MouseEvent evt) {
            jTable1MouseClicked(evt);
```

```
        }
    });
    jScrollPane1.setViewportView(jTable1);
    jLabel11.setText("Address");
    jLabel12.setText("Phone");
    txtaddress.setColumns(20);
    txtaddress.setRows(5);
    jScrollPane2.setViewportView(txtaddress);
    jButton4.setText("Cancel");
    jButton4.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton4ActionPerformed(evt);
        }
    });
 javax.swing.GroupLayout jPanel2Layout = new
javax.swing.GroupLayout(jPanel2);
    jPanel2.setLayout(jPanel2Layout);
    jPanel2Layout.setHorizontalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel2Layout.createSequentialGroup()

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.LEADING, false)
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addGap(36, 36, 36)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.TRAILING)
                        .addComponent(jLabel11)
                        .addComponent(jLabel9,
javax.swing.GroupLayout.Alignment.LEADING)
```

```
                    .addComponent(jLabel12,
javax.swing.GroupLayout.Alignment.LEADING))
                .addGap(18, 18, 18)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.LEADING, false)
                    .addComponent(jScrollPane2,
javax.swing.GroupLayout.DEFAULT_SIZE, 194, Short.MAX_VALUE)
                    .addComponent(txtphone)
                    .addComponent(txtname))
                .addGap(34, 34, 34))
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(6, 6, 6)
                .addComponent(jButton1,
javax.swing.GroupLayout.PREFERRED_SIZE, 68,
javax.swing.GroupLayout.PREFERRED_SIZE)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(jButton2,
javax.swing.GroupLayout.PREFERRED_SIZE, 64,
javax.swing.GroupLayout.PREFERRED_SIZE)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(jButton3,
javax.swing.GroupLayout.PREFERRED_SIZE, 77,
javax.swing.GroupLayout.PREFERRED_SIZE)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(jButton4, javax.swing.GroupLayout.DEFAULT_SIZE,
78, Short.MAX_VALUE)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)))
            .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE,
623, Short.MAX_VALUE)
            .addContainerGap())
```

```
    );
    jPanel2Layout.setVerticalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel2Layout.createSequentialGroup()
            .addGap(21, 21, 21)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                .addComponent(txtname, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel9))
            .addGap(28, 28, 28)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.LEADING)
                .addComponent(jLabel11)
                .addComponent(jScrollPane2,
javax.swing.GroupLayout.PREFERRED_SIZE, 55,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(16, 16, 16)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                .addComponent(txtphone,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel12))

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.LEADING)
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addGap(31, 31, 31)
```

```
.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                    .addComponent(jButton2,
javax.swing.GroupLayout.PREFERRED_SIZE, 37,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jButton1,
javax.swing.GroupLayout.PREFERRED_SIZE, 35,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jButton3,
javax.swing.GroupLayout.PREFERRED_SIZE, 37,
javax.swing.GroupLayout.PREFERRED_SIZE)))
              .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanel2Layout.createSequentialGroup()
                  .addGap(30, 30, 30)
                  .addComponent(jButton4,
javax.swing.GroupLayout.PREFERRED_SIZE, 38,
javax.swing.GroupLayout.PREFERRED_SIZE)))
          .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
       .addGroup(jPanel2Layout.createSequentialGroup()
          .addComponent(jScrollPane1,
javax.swing.GroupLayout.PREFERRED_SIZE, 260,
javax.swing.GroupLayout.PREFERRED_SIZE)
          .addGap(0, 14, Short.MAX_VALUE))
    );
 jLabel1.setFont(new java.awt.Font("Tahoma", 1, 36)); // NOI18N
    jLabel1.setText("Author");
    javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
       layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
       .addGroup(layout.createSequentialGroup()
          .addContainerGap()
```

```
            .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addContainerGap())
        .addGroup(layout.createSequentialGroup()
            .addGap(360, 360, 360)
            .addComponent(jLabel1)
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()
            .addGap(29, 29, 29)
            .addComponent(jLabel1)
            .addGap(18, 18, 18)
            .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(21, Short.MAX_VALUE))
    );
    }
}
}
```



4. Book Module for library management system

In this module, we can edit the details of the book. Name the file as **book.java**.

```
package library;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.SQLException;
import java.util.Vector;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;
public class Book extends javax.swing.JFrame {
    public Book() {
        initComponents();
        Connect();
        Author();
        Category();
        Publisher();
        table_update();
    }
   Connection con;
    PreparedStatement pst;
      public class CategoryItem
{
int id;
String name;
```

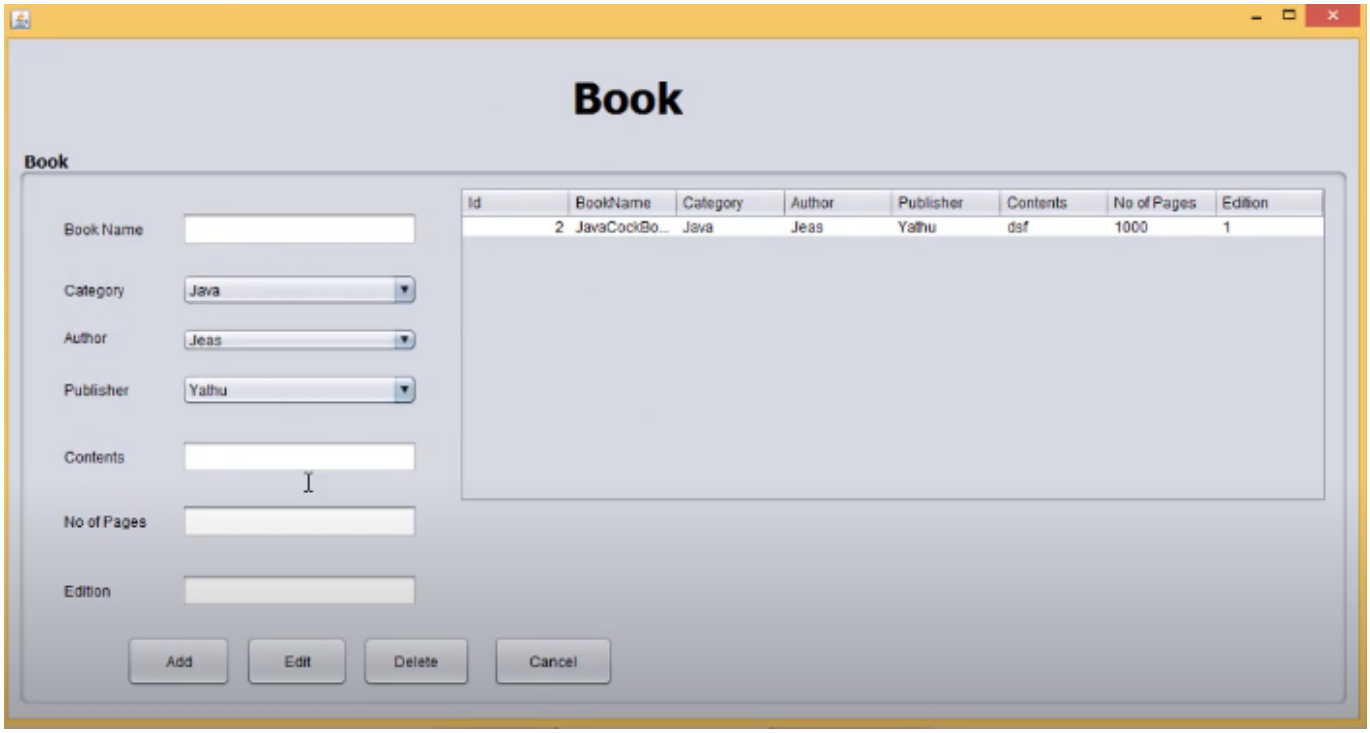
```
public CategoryItem( int id, String name )
{
this.id = id;
this.name = name;
}
public String toString()
{
return name;
}
}
  public void Connect()
    {
       try {
          Class.forName("com.mysql.jdbc.Driver");
            con = DriverManager.getConnection("jdbc:mysql://localhost/library","root","");
       } catch (ClassNotFoundException ex) {
          Logger.getLogger(Publisher.class.getName()).log(Level.SEVERE, null, ex);
       } catch (SQLException ex) {
          Logger.getLogger(Publisher.class.getName()).log(Level.SEVERE, null, ex);
      }
private void Author()
    {
          pst = con.prepareStatement("select * from author");
          ResultSet rs = pst.executeQuery();
          txtauthor.removeAllItems();
          while(rs.next())
          {
             txtauthor.addItem(new Authoritem(rs.getInt(1),rs.getString(2)) );
          }
       }
       catch (SQLException ex) {
          Logger.getLogger(Book.class.getName()).log(Level.SEVERE, null, ex);
       }
    }
   private void Publisher()
    {
       try {
          pst = con.prepareStatement("select * from publisher");
          ResultSet rs = pst.executeQuery();
          txtpub.removeAllItems();
          while(rs.next())
          {
             txtpub.addItem(new Publisheritem(rs.getInt(1),rs.getString(2)) );
          }
       }  catch (SQLException ex) {
          Logger.getLogger(Book.class.getName()).log(Level.SEVERE, null, ex);
       }
    }
   private void table_update()
    {
    int c;
          try {
```

```
            pst = con.prepareStatement("SELECT
book.id,book.bookname,c.category,a.name,p.name,book.contents,book.nopages,book.edition FROM books book
JOIN category c ON book.cat_id = c.id JOIN author a ON book.author_id = a.id JOIN publisher p ON book.p_id =
p.id");
            ResultSet rs = pst.executeQuery();
            ResultSetMetaData rsd = rs.getMetaData();
            c = rsd.getColumnCount();
            DefaultTableModel d = (DefaultTableModel)jTable1.getModel();
            d.setRowCount(0);
            while(rs.next())
            {
               Vector v2 = new Vector();
               for(int i=1; i<=c; i++)
               {
                  v2.add(rs.getString("book.id"));
                  v2.add(rs.getString("book.bookname"));
                    v2.add(rs.getString("c.category"));
                  v2.add(rs.getString("a.name"));
                  v2.add(rs.getString("p.name"));
                  v2.add(rs.getString("book.contents"));
                  v2.add(rs.getString("book.nopages"));
                  v2.add(rs.getString("book.edition"));
               }
               d.addRow(v2);
            }
      } catch (SQLException ex) {
         Logger.getLogger(author.class.getName()).log(Level.SEVERE, null, ex);
      }
   }
  private void initComponents() {
      jPanel2 = new javax.swing.JPanel();
      jLabel9 = new javax.swing.JLabel();
      txtname = new javax.swing.JTextField();
      jButton1 = new javax.swing.JButton();
      jButton2 = new javax.swing.JButton();
      jButton3 = new javax.swing.JButton();
      jScrollPane1 = new javax.swing.JScrollPane();
      jTable1 = new javax.swing.JTable();
      jLabel11 = new javax.swing.JLabel();
      jLabel12 = new javax.swing.JLabel();
      txtcontents = new javax.swing.JTextField();
      jLabel13 = new javax.swing.JLabel();
      jLabel14 = new javax.swing.JLabel();
      txtcat = new javax.swing.JComboBox();
      txtauthor = new javax.swing.JComboBox();
      txtpub = new javax.swing.JComboBox();
      jLabel15 = new javax.swing.JLabel();
      txtpages = new javax.swing.JTextField();
      jLabel16 = new javax.swing.JLabel();
      txtpeditor = new javax.swing.JTextField();
      jButton4 = new javax.swing.JButton();
      jLabel1  = new javax. swing.JLabel();
```

```
    javax.swing.GroupLayout jPanel2Layout = new javax.swing.GroupLayout(jPanel2);
        jPanel2.setLayout(jPanel2Layout);
        jPanel2Layout.setHorizontalGroup(
            jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addGroup(jPanel2Layout.createSequentialGroup()
                        .addGap(26, 26, 26)
                        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(jLabel11)
                            .addComponent(jLabel9)
                            .addComponent(jLabel13)
                            .addComponent(jLabel14)
                            .addComponent(jLabel15)
                            .addComponent(jLabel12)
                            .addComponent(jLabel16))
                            .addGap(28, 28, 28)
                        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING,
false)
                            .addComponent(txtpeditor, javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(txtcontents, javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(txtname, javax.swing.GroupLayout.Alignment.LEADING)
                            .addComponent(txtcat, javax.swing.GroupLayout.Alignment.LEADING, 0,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                            .addComponent(txtauthor, javax.swing.GroupLayout.Alignment.LEADING, 0,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                            .addComponent(txtpub, javax.swing.GroupLayout.Alignment.LEADING, 0, 194,
Short.MAX_VALUE)
                            .addComponent(txtpages, javax.swing.GroupLayout.Alignment.LEADING))
                        .addGap(34, 34, 34)
                        .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 713,
Short.MAX_VALUE))
                    .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(77, 77, 77)
                        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 86,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                        .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 84,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                        .addComponent(jButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 89,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(18, 18, 18)
                        .addComponent(jButton4, javax.swing.GroupLayout.PREFERRED_SIZE, 99,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(0, 0, Short.MAX_VALUE)))
                .addContainerGap())
        );
    jPanel2Layout.setVerticalGroup(
            jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addGap(21, 21, 21)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(txtname, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jLabel9))
                    .addGap(23, 23, 23)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jLabel11)
                        .addComponent(txtcat, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                    .addGap(18, 18, 18)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .addComponent(jLabel13)
                        .addComponent(txtauthor, javax.swing.GroupLayout.PREFERRED_SIZE, 20,
javax.swing.GroupLayout.PREFERRED_SIZE))
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .addGroup(jPanel2Layout.createSequentialGroup()
                            .addGap(23, 23, 23)
                            .addComponent(jLabel14))
                        .addGroup(jPanel2Layout.createSequentialGroup()
                            .addGap(18, 18, 18)
                            .addComponent(txtpub, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
                        swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)))
                    .addGap(28, 28, 28)
                .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(txtcontents, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jLabel15))
                    .addGap(25, 25, 25)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jLabel12)
                        .addComponent(txtpages, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))          .addGap(29,
29, 29)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jLabel16)
                        .addComponent(txtpeditor, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 260,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 23, Short.MAX_VALUE)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                .addComponent(jButton1, javax.swing.GroupLayout.DEFAULT_SIZE, 42, Short.MAX_VALUE)
                .addComponent(jButton2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jButton3, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jButton4, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
```

```
            .addContainerGap())
        );
     jLabel1.setFont(new java.awt.Font("Tahoma", 1, 36));
        jLabel1.setText("Book");
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
           layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
           .addGroup(layout.createSequentialGroup()
              .addGap(463, 463, 463)
              .addComponent(jLabel1)
              .addContainerGap(541, Short.MAX_VALUE))
           .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
              .addGroup(layout.createSequentialGroup()
                 .addContainerGap()
                 .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                          .addContainerGap())))
        );
     layout.setVerticalGroup(
           layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
           .addGroup(layout.createSequentialGroup()
              .addGap(26, 26, 26)
              .addComponent(jLabel1)
              .addContainerGap(437, Short.MAX_VALUE))
           .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
              .addGroup(layout.createSequentialGroup()
                 .addGap(92, 92, 92)
                 .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                 .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)))
        );
        pack();
        setLocationRelativeTo(null);
    }
    }
}
```



.
.

5. Issue Book Module in simple library management system project. In this module, we can issue books to the members. Name the file as **issue.java**.

```
package library;
import com.sun.glass.events.KeyEvent;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.SQLException;
import java.text.SimpleDateFormat;
import java.util.Vector;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;
```

```
public class Lend extends  javax. swing.JFrame
 {
    public Issue() {
        initComponents();
        Connect();
        Book();
        table_update();
    }
     Connection con;
    PreparedStatement pst;
      public class BookItem
{
int id;
String name;
public BookItem( int id, String name )
{
this.id = id;
this.name = name;
}
public String toString()
{
return name;
}
}
   public void Connect()
    {
```

```
        try {
            Class.forName("com.mysql.jdbc.Driver");
             con =
    DriverManager.getConnection("jdbc:mysql://localhost/library","root","");
        } catch (ClassNotFoundException ex) {
            Logger.getLogger(Publisher.class.getName()).log(Level.SEVERE,
    null, ex);
        } catch (SQLException ex) {
            Logger.getLogger(Publisher.class.getName()).log(Level.SEVERE,
    null, ex);
      }}
  } private void initComponents() {
        jPanel2 = new javax.swing.JPanel();
        jLabel9 = new javax.swing.JLabel();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTable1 = new javax.swing.JTable();
        jLabel11 = new javax.swing.JLabel();
        jLabel13 = new javax.swing.JLabel();
        jLabel14 = new javax.swing.JLabel();
        txtbook = new javax.swing.JComboBox();
        txtid = new javax.swing.JTextField();
        txttdate = new com.toedter.calendar.JDateChooser();
        txtrdate = new com.toedter.calendar.JDateChooser();
        txtname = new javax.swing.JTextField();
        jLabel10 = new javax.swing.JLabel();
        jLabel1 = new javax.swing.JLabel();
        jButton1 = new javax.swing.JButton();
```

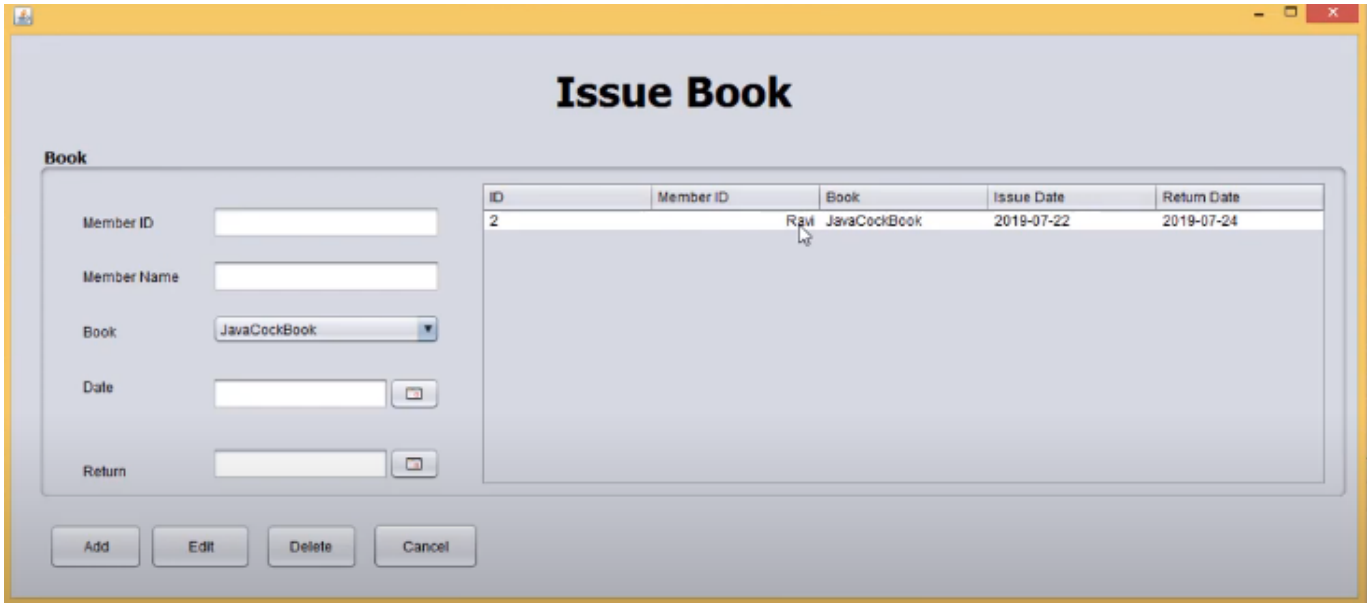
```
        jButton2 = new javax.swing.JButton();
        jButton3 = new javax.swing.JButton();
    jButton4 = new javax.swing.JButton();
        public Class getColumnClass(int columnIndex) {
                return types [columnIndex];
        }
       });
      jTable1.setCursor(new
java.awt.Cursor(java.awt.Cursor.DEFAULT_CURSOR));
      jTable1.setGridColor(new java.awt.Color(255, 255, 255));
      jTable1.addMouseListener(new java.awt.event.MouseAdapter() {
        public void mouseClicked(java.awt.event.MouseEvent evt) {
          jTable1MouseClicked(evt);
        }
     });
   jScrollPane1.setViewportView(jTable1);
     jLabel11.setText("Book");
     jLabel13.setText("Date");
     jLabel14.setText("Return");
     txtid.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
          txtidActionPerformed(evt);
        }
      });
   txtid.addKeyListener(new java.awt.event.KeyAdapter() {
        public void keyPressed(java.awt.event.KeyEvent evt) {
          txtidKeyPressed(evt);
        }
      });
   jLabel10.setText("Member Name");
     javax.swing.GroupLayout jPanel2Layout = new javax.swing.GroupLayout(jPanel2);
     jPanel2.setLayout(jPanel2Layout);
     jPanel2Layout.setHorizontalGroup(
        jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel2Layout.createSequentialGroup()
          .addGap(26, 26, 26)
          .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jLabel9)
            .addComponent(jLabel13)
            .addComponent(jLabel14)
            .addComponent(jLabel11)
            .addComponent(jLabel10))
          .addGap(28, 28, 28)
          .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING, false)
```

```
                .addComponent(txtid)
                .addComponent(txttdate, javax.swing.GroupLayout.DEFAULT_SIZE, 194, Short.MAX_VALUE)
                .addComponent(txtrdate, javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(txtname)
                .addComponent(txtbook, 0, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addGap(34, 34, 34)
            .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 720, Short.MAX_VALUE)
            .addContainerGap())
      );
    jPanel2Layout.setVerticalGroup(
        jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel2Layout.createSequentialGroup()
            .addGap(21, 21, 21)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel9)
                .addComponent(txtid, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(18, 18, 18)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(txtname, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel10))
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jLabel11)
                    .addGap(31, 31, 31))
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addGap(18, 18, 18)
                    .addComponent(txtbook)
                    .addGap(28, 28, 28)))
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jLabel13)
                .addComponent(txttdate, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(32, 32, 32)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING)
                .addComponent(jLabel14)
                .addComponent(txtrdate, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(243, 243, 243))
        .addGroup(jPanel2Layout.createSequentialGroup()
            .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 260,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(0, 0, Short.MAX_VALUE))
      );
    jLabel1.setFont(new java.awt.Font("Tahoma", 1, 36)); // NOI18N
      jLabel1.setText("Issue Book");
      jButton1.setText("Add");
      jButton1.addActionListener(new java.awt.event.ActionListener() {
```

```
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton1ActionPerformed(evt);
        }
    });
    jButton2.setText("Edit");
    jButton2.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton2ActionPerformed(evt);
        }
    });
    jButton3.setText("Delete");
    jButton3.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton3ActionPerformed(evt);
        }
    });
    jButton4.setText("Cancel");
    jButton4.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton4ActionPerformed(evt);
        }
    });
  javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGap(21, 21, 21)
                    .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                .addGroup(layout.createSequentialGroup()
                    .addGap(463, 463, 463)
                    .addComponent(jLabel1)
                    .addGap(0, 0, Short.MAX_VALUE)))
            .addContainerGap())
        .addGroup(layout.createSequentialGroup()
            .addGap(32, 32, 32)
            .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 80,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 86,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
            .addComponent(jButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 79,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
            .addComponent(jButton4, javax.swing.GroupLayout.PREFERRED_SIZE, 91,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
    );
```

```
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
            .addGap(26, 26, 26)
            .addComponent(jLabel1)
            .addGap(26, 26, 26)
            .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE, 291,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jButton3, javax.swing.GroupLayout.PREFERRED_SIZE, 39,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 39,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 39,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jButton4, javax.swing.GroupLayout.DEFAULT_SIZE, 40, Short.MAX_VALUE))
            .addGap(24, 24, 24))
    );
}

private void txtidKeyPressed(java.awt.event.KeyEvent evt) {
    if(evt.getKeyCode() == KeyEvent.VK_ENTER)
    {
        String id = txtid.getText();
        try {
            pst = con.prepareStatement("select * from member where id=?");
            pst.setString(1, id);
            ResultSet rs = pst.executeQuery();
            if(rs.next()==false)
            {
                JOptionPane.showMessageDialog(this,"Member ID not Found");
            }
            else
            {
                String productname = rs.getString("name");
                txtname.setText(productname.trim());
            }
        } catch (SQLException ex) {
            Logger.getLogger(Lend.class.getName()).log(Level.SEVERE, null, ex);
        }
    }
  }
```



Return Book Module in library management system

In this module, we can return the book to the members. Name the file **return.java**.

```
package library;
import java.awt.event.KeyEvent;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.SQLException;
import java.util.Vector;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;
public class Return extends javax.swing.JFrame {
```

```
public Return() {
    initComponents();
     Connect();
     table_update();
}
Connection con;
PreparedStatement pst;
@SuppressWarnings("unchecked")
private void initComponents() {
    jPanel2 = new javax.swing.JPanel();
    jLabel9 = new javax.swing.JLabel();
    jScrollPane1 = new javax.swing.JScrollPane();
    jTable1 = new javax.swing.JTable();
    jLabel11 = new javax.swing.JLabel();
    jLabel13 = new javax.swing.JLabel();
    jLabel14 = new javax.swing.JLabel();
    txtid = new javax.swing.JTextField();
    jLabel10 = new javax.swing.JLabel();
    jLabel1 = new javax.swing.JLabel();
    jLabel2 = new javax.swing.JLabel();
    txtdate = new javax.swing.JTextField();
    jLabel5 = new javax.swing.JLabel();
    txtelp = new javax.swing.JTextField();
    txtfine = new javax.swing.JTextField();
    jButton1 = new javax.swing.JButton();
    jLabel3 = new javax.swing.JLabel();
    jButton2 = new javax.swing.JButton();
    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    jPanel2.setBorder(javax.swing.BorderFactory.createTitledBorder(null, "Return",
javax.swing.border.TitledBorder.DEFAULT_JUSTIFICATION,
```

```
javax.swing.border.TitledBorder.DEFAULT_POSITION, new
java.awt.Font("Tahoma", 1, 14))); // NOI18N
    jLabel9.setText("Member ID");
    jTable1.setModel(new javax.swing.table.DefaultTableModel(
        new Object [][] {
   },
        new String [] {
            "ID", "Member ID", "Member Name", "Book", "Return Date", "Days Elp",
"Fine"
        }
    ) {
        Class[] types = new Class [] {
            java.lang.String.class, java.lang.Integer.class, java.lang.String.class,
java.lang.String.class, java.lang.String.class, java.lang.String.class,
java.lang.String.class
        };
        public Class getColumnClass(int columnIndex) {
            return types [columnIndex];
        }
    });
    jTable1.setCursor(new java.awt.Cursor(java.awt.Cursor.DEFAULT_CURSOR));
    jTable1.setGridColor(new java.awt.Color(255, 255, 255));
    jTable1.addMouseListener(new java.awt.event.MouseAdapter() {
        public void mouseClicked(java.awt.event.MouseEvent evt) {
            jTable1MouseClicked(evt);
        }
    });
 jScrollPane1.setViewportView(jTable1);
    jLabel11.setText("Book");
    jLabel13.setText("Return Date");
    jLabel14.setText("Fine");
```

```
        txtid.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                txtidActionPerformed(evt);
            }
        });
        txtid.addKeyListener(new java.awt.event.KeyAdapter() {
            public void keyPressed(java.awt.event.KeyEvent evt) {
                txtidKeyPressed(evt);
            }
        });
     jLabel10.setText("Member Name");
        jLabel1.setFont(new java.awt.Font("Tahoma", 1, 15)); // NOI18N
        jLabel1.setText("Membername");
        jLabel2.setFont(new java.awt.Font("Tahoma", 1, 15)); // NOI18N
        jLabel2.setText("Book");
        jLabel5.setText("Days Elapsed");
        javax.swing.GroupLayout jPanel2Layout = new
    javax.swing.GroupLayout(jPanel2);
        jPanel2.setLayout(jPanel2Layout);
        jPanel2Layout.setHorizontalGroup(

    jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(26,26,26)

    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
    .LEADING)
                    .addComponent(jLabel9)
                    .addComponent(jLabel13)
                    .addComponent(jLabel14)
                    .addComponent(jLabel11)
```

```
                .addComponent(jLabel10)
                .addComponent(jLabel5))
            .addGap(28, 28, 28)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.LEADING, false)
                .addComponent(txtid, javax.swing.GroupLayout.Alignment.TRAILING)
                .addComponent(jLabel1)
                .addComponent(jLabel2)
                .addComponent(txtdate, javax.swing.GroupLayout.DEFAULT_SIZE, 194,
Short.MAX_VALUE)
                .addComponent(txtelp))
            .addGap(34, 34, 34)
            .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE,
740, Short.MAX_VALUE)
            .addContainerGap())
    );
 jPanel2Layout.setVerticalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel2Layout.createSequentialGroup()

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.TRAILING)
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addGap(21, 21, 21)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                        .addComponent(jLabel9)
                        .addComponent(txtid,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
```

```
                .addGap(18, 18, 18)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                    .addComponent(jLabel10)
                    .addComponent(jLabel1))
                .addGap(18, 18, 18)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                    .addComponent(jLabel11)
                    .addComponent(jLabel2))
                        .addGap(25, 25, 25)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                    .addComponent(jLabel13)
                    .addComponent(txtdate,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

.addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment
.BASELINE)
                    .addComponent(jLabel5)
                    .addComponent(txtelp,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(40, 40, 40)
                .addComponent(jLabel14))
            .addComponent(jScrollPane1,
```

```
javax.swing.GroupLayout.PREFERRED_SIZE, 260,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(0, 0, Short.MAX_VALUE))
    );
 txtfine.setText("jTextField1");
    jButton1.setText("Return");
    jButton1.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton1ActionPerformed(evt);
        }
    });
    jLabel3.setFont(new java.awt.Font("Tahoma", 1, 36)); // NOI18N
    jLabel3.setText("Return");
    jButton2.setText("Cancel");
    jButton2.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            jButton2ActionPerformed(evt);
        }
    });
 javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()

.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADI
NG)
                .addGroup(layout.createSequentialGroup()
                    .addGap(148, 148, 148)
                    .addComponent(txtfine,
```

```
javax.swing.GroupLayout.PREFERRED_SIZE, 186,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGroup(layout.createSequentialGroup()
                    .addGap(89, 89, 89)
                    .addComponent(jButton1,
javax.swing.GroupLayout.PREFERRED_SIZE, 99,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(18, 18, 18)
                    .addComponent(jButton2,
javax.swing.GroupLayout.PREFERRED_SIZE, 112,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGroup(layout.createSequentialGroup()
                    .addGap(493, 493, 493)
                    .addComponent(jLabel3)))
            .addContainerGap(515, Short.MAX_VALUE))

.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADI
NG)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addContainerGap()))
    );
 layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()
            .addGap(31, 31, 31)
            .addComponent(jLabel3)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 274,
Short.MAX_VALUE)
```

```
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADI
NG)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()
                .addComponent(txtfine,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(117, 117, 117))
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()

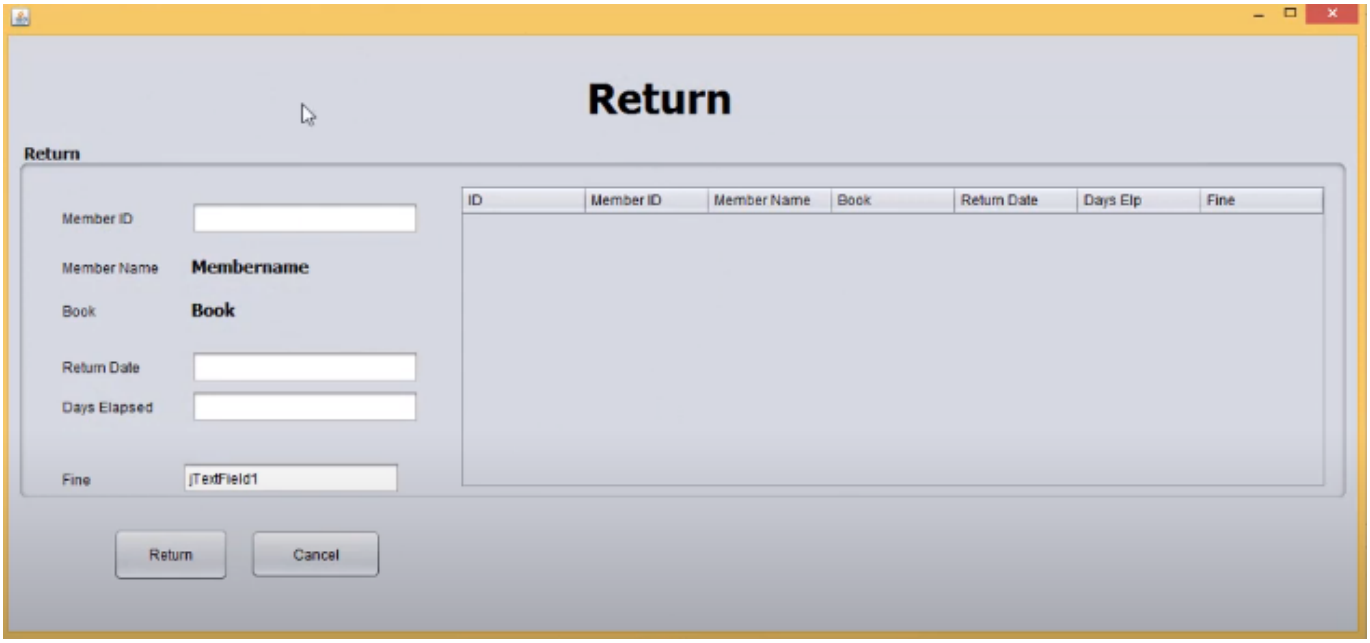
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELI
NE)
                    .addComponent(jButton1,
javax.swing.GroupLayout.PREFERRED_SIZE, 45,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jButton2,
javax.swing.GroupLayout.PREFERRED_SIZE, 42,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(43, 43, 43))))

.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADI
NG)
        .addGroup(layout.createSequentialGroup()
            .addGap(92, 92, 92)
            .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(108, Short.MAX_VALUE)))
    );
    pack();
        setLocationRelativeTo(null);
}
```

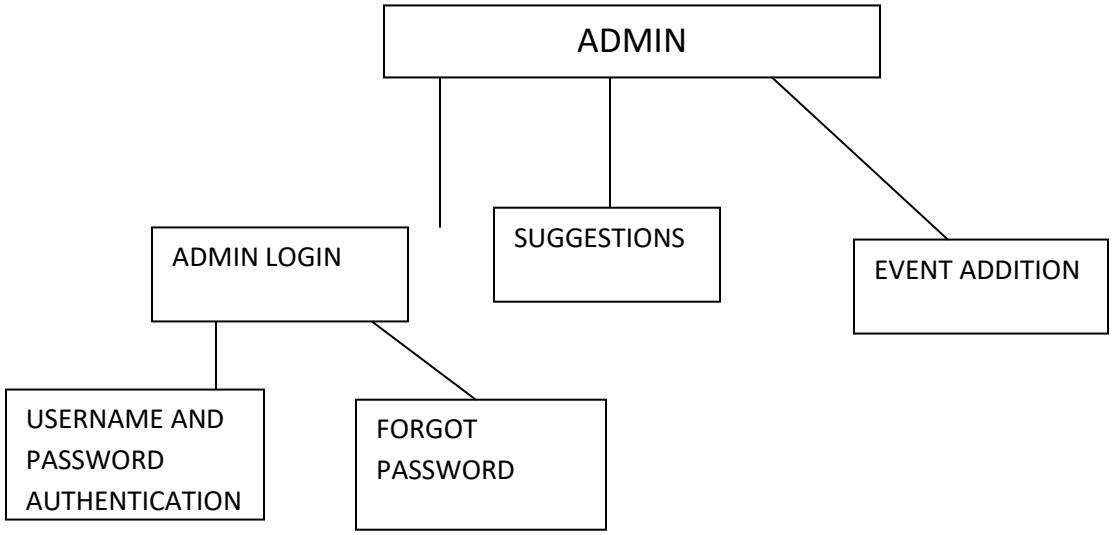
```
  public void Connect()
{
    try {
        Class.forName("com.mysql.jdbc.Driver");
          con = DriverManager.getConnection("jdbc:mysql://localhost/slibr","root","");
    } catch (ClassNotFoundException ex) {
        Logger.getLogger(Publisher.class.getName()).log(Level.SEVERE, null, ex);
    } catch (SQLException ex) {
        Logger.getLogger(Publisher.class.getName()).log(Level.SEVERE, null, ex);
    }
private void txtidKeyPressed(java.awt.event.KeyEvent evt) {//GEN-
FIRST:event_txtidKeyPressed
   if(evt.getKeyCode() == KeyEvent.VK_ENTER)
    {
        String id = txtid.getText();
        try {
            pst = con.prepareStatement("SELECT
m.name,b.bookname,l.returndate,DATEDIFF(NOW(),l.returndate) as elap FROM
lend l JOIN books b ON l.bookid = b.id JOIN member m ON l.memberid = m.id and
l.memberid = ?");
              pst.setString(1, id);
            ResultSet rs = pst.executeQuery();
            if(rs.next()==false)
            {
                JOptionPane.showMessageDialog(this,"Member ID not Found");
            }
            else
            {
                String mname = rs.getString("m.name");
                 String bname = rs.getString("b.bookname");
                  jLabel1.setText(mname.trim());
```

```
                jLabel2.setText(bname.trim());
                 String date = rs.getString("l.returndate");
                  String elp = rs.getString("elap");

                 int elaped = Integer.parseInt(elp);
                 if(elaped > 0)
                 {
                    txtelp.setText(elp);
                    int fine = elaped * 50;
                    txtfine.setText(String.valueOf(fine));
                 }
          else
                 {
                    txtelp.setText("0");
                    txtfine.setText("0");
                 }
                 txtdate.setText(date.trim());
            }
        } catch (SQLException ex) {
            Logger.getLogger(Return.class.getName()).log(Level.SEVERE, null, ex);
        }
     }
    }
```

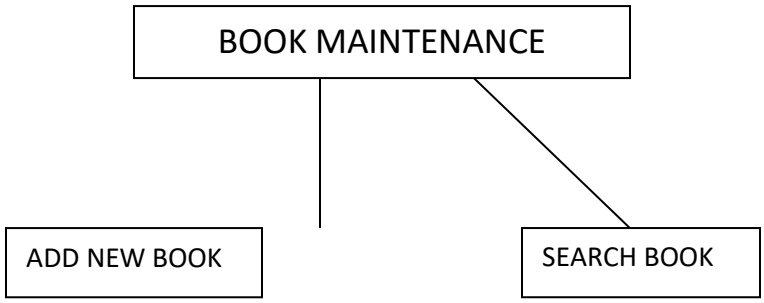

Return
Return
Member ID
Member Name
Membername
Book
Book
Return Date
Days Elapsed
Fine
jTextField1
Return
Cancel
ID
Member ID
Member Name
Book
Return Date
Days Elp
Fine

- **ADMIN Module**



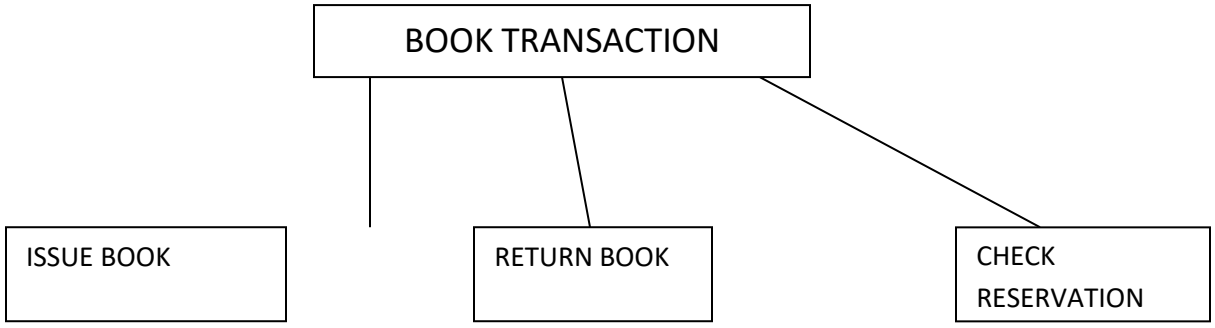
The following module contains various facilities like teacher login, suggestions, and event addition. Further any teacher if at any moment forgets his/her password he/she can retrieve it from ‘forgot password’ option.

- **4.1.4 Book Module**



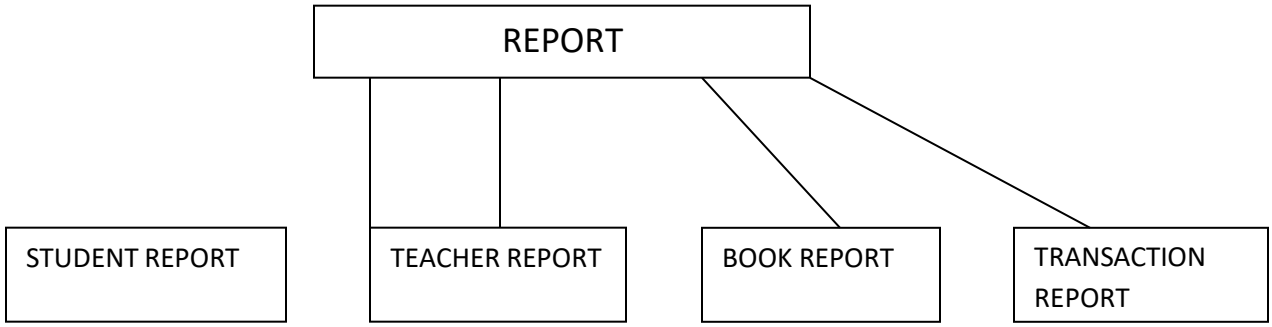
The following module contains various facilities like add new book and search book. In the 'add new book' section if any new book comes in the library then the librarian can add its specifications. Similarly if the user wants to search for a specific book then he/she can use search book option to do it.

- **4.1.5 Book Transaction Module**



The following module contains various facilities like issue book, return book and check reservation.

- **4.1.6 Report Module**



The following module contains various facilities like student report, teacher report, book report, and transaction report.

SYSTEM TESTING

The aim of the system testing process was to determine all defects in our project .The program was subjected to a set of test inputs and various observations were made and based on these observations it will be decided whether the program behaves as expected or not.

Our Project went through two levels of testing

1.Unit testing

2.integration testing

UNIT TESTING

Unit testing is undertaken when a module has been created and succesfully reviewed .In order to test a single module we need to provide a complete environment ie besides the module we would require

- The procedures belonging to other modules that the module under test calls
- Non local data structures that module accesses
- A procedure to call the functions of the module under test with appropriate parameters

Unit testing was done on each and every module that is described under module description

1. Test For the admin module

- Testing admin login form-This form is used for log in of administrator of the system.In this we enter the username and password if both are correct administration page will open other wise if any of data is wrong it will get redirected back to the login page and again ask for username and password
- Student account addition- In this section the admin can verify student details from student academinc info and then only add student details to main library database it contains add and delete buttons if user click add button data will be added to student database and if he clicks delete button the student data will be deleted
- Book Addition- Admin can enter details of book and can add the details to the main book table also he can view the books requests .

2. Test for Student login module
   - Test for Student login Form-This form is used for log in of Student .In this we enter thelibraryid, username and password if all these are correct student login page will open other wise if any of data is wrong it will get redirected back to the login page and again ask for libraryid, username and password.
   - Test for account creation- This form is used for new account creation when student does not fill the form completely it asks again to fill the whole form when he fills the form fully it gets redirected to page which show waiting forconformation message as his data will be only added by administrator after verification.
3. Test for teacher login module-
   - Test for teacher login form- This form is used for log in of teacher .In this we enter the username and password if all these are correct teacher login page will open otherwise if any of data is wrong it will get redirected back to the login page and again ask for username and password.

INTEGRATION TESTING

In this type of testing, we test various integration of the project module by providing the input . The primary objective is to test the module interfaces in order to ensure that no errors areoccurring when one module invokes the other module.

CONCLUSION & FUTURE SCOPE

This website provides a computerized version of library management system which will benefit the students as well as the staff of the library.
It makes entire process online where student can search books, staff can generate reports and do book transactions. It also has a facility for ADMIN login where ADMIN can login and can see status of books issued.
In this project, we developed a GUI-based project, a *Library Management System Project in Java and MySQL* or in other words *book library management system code in java*. The users are able to perform the operations such as *Login, View Categories, Book details, Author details, Book issue, and Book return*.

There is a future scope of this facility that many more features such as Teacher, Student modules can be added to this project thus making it more interactive more user friendly and project which fulfills each users need in the best way possible

REFERENCES

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